COURSE DESCRIPTION			
Program	**Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Civil Engineering (Construction Technology)**		
Course Title	**Fundamentals of Civil Engineering**		
Course Code	**1011**	**Semester**	**I**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:0:3:0**	**Credits**	**4.5**
CIE Marks	**40**	**ESE Marks**	**100**

Introduction

The Fundamentals of Civil Engineering introduces the basic concepts of building planning and regulations, building components, construction practices and essential building services. Additionally, it provides hands-on experience to comprehend and implement these ideas in basic construction projects.

Course Objectives

1	To impart basic principles of building construction, planning, and regulations
2	To familiarize the masonry and finishing works used in building construction
3	To familiarize the various components of buildings
4	To gain practical knowledge of various building services

Course Pre-requisite

Topic	Course code	Course Title	Semester
NIL			

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply fundamental civil engineering principles in surveying, area measurement, regulations and building planning.	Apply
CO2	Practice appropriate masonry and finishing techniques for durable and quality building construction	Apply
CO3	Apply knowledge of building components, foundations, and roofing materials in various construction scenarios	Apply
CO4	Demonstrate basic building service concepts related to plumbing, electrical and fire safety in buildings.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	3	3	-	-	-	-	-	-
CO2	3	2	-	-	3	-	-	-	-	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	3	-	-	-	-	-	-
Total	12	6	-	3	9	-	-	-	-	-	-
Correlation Level	3	2	-	3	3	-	-	-	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	40	60	140

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Fundamentals of Civil Engineering	• Introduction to Civil Engineering – Relevance of civil engineering in development of nation, Major disciplines of Civil engineering • Introduction to buildings – Types of buildings based on occupancy as per NBC, Based on type of construction - Load Bearing Structures & Framed Structures • Selection of site for buildings • Building rules and regulations – Building area concepts – Plinth area, Built up area, Floor area, Carpet area, Rules and bye-laws of sanctioning authorities for construction work as per Kerala Building Rules - coverage, floor space index, exterior and interior open air space requirements (residential building only) • Surveying - principles of surveying - purpose of surveying - Plane surveying and geodetic surveying. Classification of surveys - based on instruments,based on functions, based on nature of field, Units of measurements	11	0
Module I	Fundamentals of Civil Engineering	• Identify and classify buildings in nearby area as per NBC and tabulate the report • Compute plinth area, built up area, floor area, & carpet area of a building • Determine the distance between two intervisible survey stations using tape	0	9
Module II	Masonry and Finishes	• Brick masonry - Terms used in brick masonry-Bonds in brick masonry- header bond, stretcher bond, english bond, flemish bond. Requirements of Brick Masonry • Stone masonry - Types - Rubble masonry, Ashlar Masonry • Pointing - Necessity - Types of pointing - flush, recessed, weathered, keyed or grooved pointing • Plastering - Definition- Objectives - Requirements of a good plaster • Painting – Necessity, Ingredients, Characteristics of an Ideal paint, Types of paint	12	0
Module II	Masonry and Finishes	• Construct one brick thick & one and a half brick thick wall corner in English Bond • Construct one brick thick & one and a half brick thick wall corner in Flemish Bond • Construct one brick thick pier in English bond and one and a half brick thick pier in English Bond and Flemish bond • Perform painting on any surface with enamel and emulsion paints	0	12
Module III	Introduction to Building components	• Basic components of a building and their functions– Substructure -Foundation, plinth, Superstructure - wall, lintel, sunshade, floor, roof, parapet • Foundation and its purpose, Types of foundations – shallow and deep, Shallow foundation- Wall Footing, Column Footing, Combined Footing, Strap footing, Grillage Foundation, Raft Foundation • Deep foundation- Pile Foundation, Well foundation (Introduction only) • Roofs - Roofing Materials - RCC, MP Tiles, Ceramic roofing tiles, Thatched, G.I. sheets, Corrugated G.I. Sheets, Plastic and Fiber Sheets	12	0
Module III	Introduction to Building components	• Prepare a neatly labeled sketch of a building showing all components • Draw plan and sectional views of shallow foundation- isolated footing, combined footing and raft foundation • Identify and prepare a report on different types of innovative roofing materials relevant for building construction and compare their features	0	9
Module IV	Building services and Fire Safety	• Introduction to plumbing, Components of plumbing system- Pipes, fittings & fixtures • Sanitary pipes, fittings and fixtures • Basic electrical terms: Load, Circuit, Phase, Neutral, Earth. • Difference between Single Phase and Three Phase Power Supply (Basic Concept) • Structure of a domestic electrical supply system, Basic components for electric connection: Service Connection, Energy meter, Main switch • Protection Devices in Domestic Wiring: Fuse (rewirable and cartridge), Miniature Circuit Breaker, Earth leakage circuit breaker, (Purpose only) • Wiring in residential buildings:Types of Wiring Systems (Basic Concept only): Cleat wiring, Batten wiring, Casing and capping wiring, Conduit wiring, Wiring materials (listing only). • Fire safety requirements in a building.	10	0
Module IV	Building services and Fire Safety	• Carry out fitting of pipes and taps • Perform thread cutting on G I pipe/PVC pipes • Prepare a report on current fire safety measures and protocols followed in your college	0	9

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Relevance of civil engineering	Describe the relevance of civil engineering and its major disciplines	L	CO1	Understand	1
Classification of buildings	Classify the buildings based on occupancy as per NBC	L	CO1	Remember	1
Classification of buildings	Classify the buildings based on type of construction	L	CO1	Remember	1
Classification of buildings	Identify and classify buildings in nearby area as per NBC and tabulate the report	P	CO1	Apply	3
Site selection of building	Explain the principles of site selection for buildings	L	CO1	Understand	1
Building areas and rules	Define different types of building areas(Plinth area, Built up area, Floor area, Carpet area,)	L	CO1	Remember	1
Building areas and rules	Define Coverage & FSI	L	CO1	Understand	1
Building areas and rules	Compute plinth area, built up area, floor area, & carpet area of a building	P	CO1	Apply	3
Building areas and rules	Describe the exterior and interior open air space requirements as per KBR	L	CO1	Understand	1
Surveying	Describe surveying and its principles	L	CO1	Understand	1
Surveying	Explain the purpose of surveying	L	CO1	Understand	0.5
Surveying	Differentiate plane surveying and geodetic surveying	L	CO1	Understand	1
Surveying	Determine the distance between two intervisible survey stations using tape	P	CO1	Apply	3
Surveying	Classify surveys based on instruments, based on functions and based on nature of field	L	CO1	Understand	1.5
Module II					
Brick Masonry	Define masonry.	L	CO2	Remember	0.5
Brick Masonry	Define common terms used in brick masonry.	L	CO2	Remember	0.5
Brick Masonry	Differentiate various types of bonds in brick masonry (header bond, stretcher bond, English bond, Flemish bond.)	L	CO2	Understand	1
Brick Masonry	Construct one brick thick & one and a half brick thick wall corner in English Bond	P	CO2	Apply	3
Brick Masonry	Explain the requirements of brick masonry.	L	CO2	Understand	1
Brick Masonry	Draw the plan and elevation of one brick & one and a half brick English bond.	L	CO2	Understand	1
Brick Masonry	Draw the plan and elevation of one brick & one and a half brick Flemish bond.	L	CO2	Understand	1
Brick Masonry	Construct one brick thick & one and a half brick thick wall corner in Flemish Bond.	P	CO2	Apply	3
Stone masonry	Differentiate rubble and ashlar masonry.	L	CO2	Understand	1
Pointing	Describe the necessity of pointing	L	CO2	Understand	0.5
Pointing	Identify different types of pointing based on finish (Flush, recessed, weathered, keyed or grooved)	L	CO2	Understand	1.5
Pointing	Construct one brick thick pier in English bond and one and a half brick thick pier in English Bond and Flemish bond	P	CO2	Apply	3
Plastering	Define plastering and state its objectives	L	CO2	Remember	0.5
Plastering	Explain requirements of a good plaster	L	CO2	Understand	0.5
Painting	State the necessity of painting in buildings	L	CO2	Understand	0.5

Painting	Describe the functions of various ingredients of paint.	L	CO2	Understand	0.5
Painting	List out the characteristics of an ideal paint	L	CO2	Remember	0.5
Painting	Describe different types of paint	L	CO2	Understand	1
Painting	Perform painting on any surface with enamel and emulsion paints	P	CO2	Apply	3
Module III					
Introduction to building components	Describe the basic components of a building in substructure and their functions - Foundation, plinth	L	CO3	Understand	1
Introduction to building components	Describe the basic components of a building in superstructure and their functions - wall, lintel, sunshade, floor, roof, parapet	L	CO3	Understand	2
	Series Exam -I	P	CO2	Apply	3
Foundation	Explain Foundation and its purpose.	L	CO3	Understand	1
Foundation	Compare shallow and deep foundation	L	CO3	Understand	2
Foundation	Prepare a neatly labelled sketch of a building showing all components	P	CO3	Apply	3
Foundation	Describe various types of shallow foundation - Wall Footing, Column Footing, Combined Footing, Strap footing, Grillage Foundation, Raft Foundation	L	CO3	Understand	3
Foundation	Draw plan and sectional views of shallow foundation- isolated footing, combined footing and raft foundation	P	CO3	Apply	3
Foundation	Briefly describe various types of deep foundation - Pile Foundation, Well foundation	L	CO3	Understand	1
Roofing Materials	Explain various roofing materials - RCC, MP Tiles, Ceramic roofing tiles, Thatched, G.I. sheets, Corrugated G.I. Sheets, Plastic and Fiber Sheets	L	CO3	Understand	2
Roofing Materials	Prepare a report on different types of innovative roofing materials relevant for building construction and compare their features	P	CO3	Apply	3
Module IV					
Plumbing & sanitary fittings	Explain the components of plumbing system – pipes, fittings and fixtures	L	CO4	Understand	2
Plumbing & sanitary fittings	Describe sanitary pipes, fittings and fixtures	L	CO4	Understand	1
Plumbing & sanitary fittings	Carry out fitting of pipes and taps	P	CO4	Apply	3
Electrical Services	Define the terms used for electrical installation	L	CO4	Understand	1
Electrical Services	Differentiate between Single Phase and Three Phase Power Supply	L	CO4	Understand	1
Electrical Services	Describe basic components for electric connection: Service Connection, Energy meter, Main switch	L	CO4	Understand	1
Plumbing & sanitary fittings	Perform thread cutting on G I pipe/PVC	P	CO4	Apply	3
Protection devices	Explain the protective devices - fuse, MLCB, and ELCB	L	CO4	Understand	1
Wiring in residential buildings	Describe the types of wiring systems – cleat wiring, batten wiring, casing & capping wiring, conduit wiring	L	CO4	Understand	1
Wiring in residential buildings	List various wiring materials	L	CO4	Understand	0.5
Fire safety	Describe the fire safety requirements in a building	L	CO4	Understand	1.5
Fire safety	Prepare a report on current fire safety measures and protocols followed in your college.	P	CO4	Apply	3
	Series Exam - II	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		

1	Prepare a mind map of the major disciplines of civil engineering with real life examples. Identify type of construction in nearby buildings - load-bearing structure /framed structure. Interpret the theory concepts and complete lab record	6
2	Explore the site selection principles followed in nearby sites. Conduct a study on how principles of planning are applied in your house and prepare a report. Interpret the theory concepts and complete lab record	6
3	Read the key rules in relevant building codes. Compute plinth area and carpet area of given building plan. Interpret the theory concepts and complete lab record.	6
4	Explore videos on various instrument-based surveying. Tabulate common unit conversions. Interpret the theory concepts and complete lab record	6
Module II		
5	Pre class concept refresh. Explore videos of rubble masonry and ashlar masonry and prepare a report. Interpret the theory concepts and complete lab record.	6
6	Prepare a comparison table on different types of bond. Attend a short quiz related to brick masonry. Interpret the theory concepts and complete lab record.	6
7	List various tools used for pointing. Explore videos on pointing. Interpret the theory concepts and complete lab record.	6
8	Identify common defects in plastering, study their causes, and collect relevant pictures. Interpret the theory concepts and complete lab record.	6
Module III		
9	Identify the materials used for the construction of different building components. Explore short videos or animations on foundation construction and list the functions of foundation. Interpret the theory concepts and complete lab record.	6
10	Pre class concept refresh. Prepare a comparison table on different types of shallow foundation. Interpret the theory concepts and complete lab record.	6
11	Pre class concept refresh. Read articles related to foundation failures of structures. Interpret the theory concepts and complete lab record.	6
12	Prepare a report on application of roofing materials and their suitability on various climatic conditions. Also compare the cost of different roofing materials. Interpret the theory concepts and complete lab record.	6
Module IV		
13	Identify various building services in your house and prepare a report. Explore videos on various plumbing operations. Interpret the theory concepts and complete lab record.	6
14	List any 5 electrical fixtures used in your home. List any 5 electrical safety precautions followed at home. Interpret the theory concepts and complete lab record.	6
15	Prepare a comparison table on various electrical protective devices. Identify and list various fire safety equipment in a public building. Interpret the theory concepts and complete lab record.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Basic Civil Engineering	S. S. Bhavikatti	New Age International
2	Basic Civil Engineering	Satheesh Gopi	Pearson

REFERENCE			
Sl. No.	Title	Author	Publication
1	Building Construction	S.C Rangwala	Charotar Publishing house

2	Building Construction	S P Arora and Bindra	Dhanpat Rai Publication,Delhi
3	National Building Code	BIS	Bureau of Indian Standard, New Delhi
4	Surveying I	Punmia, B.C, Jain, Ashok Kumar; Jain, Arun Kumar	Laxmi Publications, New Delhi.
5	Laboratory Manual on Testing of Engineering Materials	Sood H.	New Age Publishers, NewDelhi
6	Building Services	S. M. Patil	Seema Publication, Mumbai
7	Plumber	G. S. Sethi	Computech Publications Ltd, New Delhi
8	Textbook of Electrical technology Part 1- Basic Electrical engineering in SI units	B L Theraja, A K Theraja	S Chand Publication

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.bis.gov.in	I
2	https://nptel.ac.in/courses/105102088	Ii & Iii
3	https://nptel.ac.in/courses/108105112	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Measuring tape	30m	6
2	Try square		6
3	Straight edge		6
4	Plumb bob		6
5	Trowel		6
6	Mortar pan		6
7	Paint brush	Conforming to IS 384	6
8	Hacksaw		6
9	Pipe wrench		6
10	Pipe vice		6
11	Die set		6
12	Die stock		6

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		15	50	40	40	50	50	60	40
Converted to		5	10	10	10	10	10		
Marks	5	5	10	10		10		60	40
Total	40							100	

Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60